CatchPoint

Let them step. Catch only when they can’t.



ME5412 Robotics for Healthcare · Team project · 3 Nov 2026

Concept render (design D3, unbuilt): Blender; mannequin Mixamo/Adobe via three.js examples; HDRI Poly Haven (CC0).

Recovery steps prevent falls — but practising them is risky



71%

of stroke patients still had impaired reactive stepping at discharge

Inness et al., Phys Ther 2014

14.7%

fell during inpatient stroke rehabilitation

Mansfield et al., Neurorehabil Neural Repair 2013

684,000

deaths from falls every year worldwide

World Health Organization

Photo: US Navy, public domain (Wikimedia Commons). Today therapists guard patients by hand, so they tend to over-support.

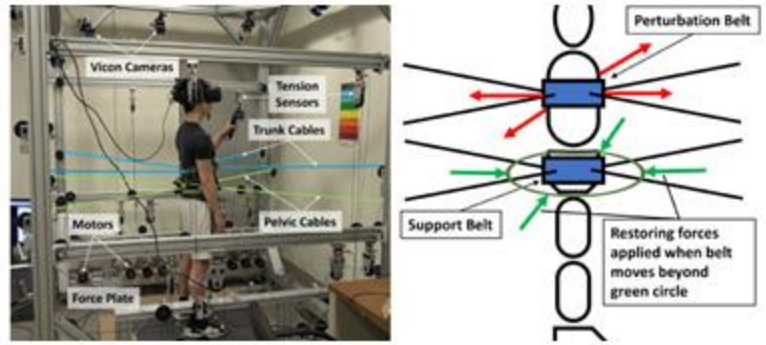
Existing solutions lock too early or too late



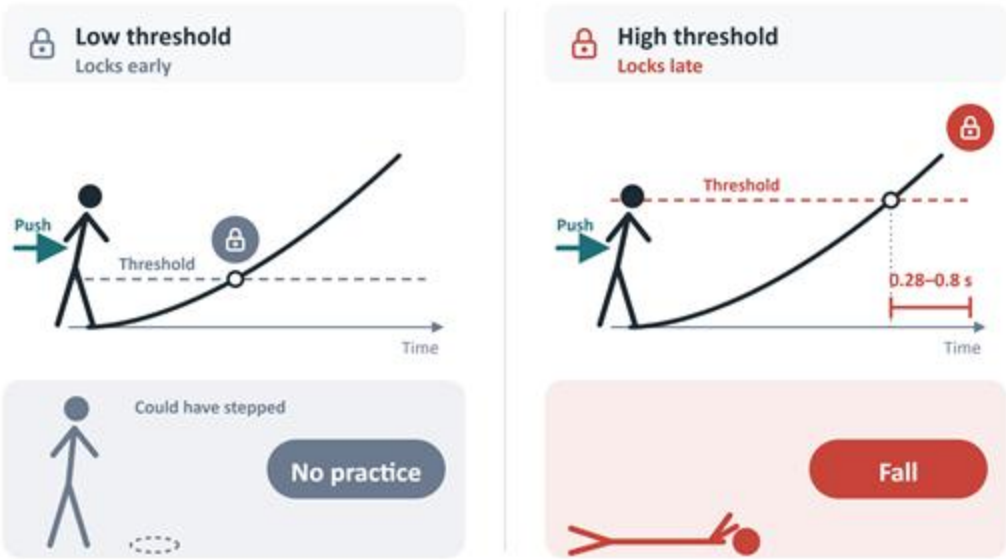
MRBA (NTU): brakes lock when a sensor threshold trips

Photo omitted in public version
(Hocoma Andago — see source link)

Andago (Hocoma): overground support walker



RobUST: pelvic force field outside a fixed boundary



Locks early — the patient never gets to practise stepping.

Locks late — response takes 0.28–0.8 s — they fall anyway.

Photos: NTU MRBA, J NeuroEng Rehabil 2023 (CC BY 4.0) · Hocoma Andago, © Hocoma/DIH · RobUST, Omofuma et al., Bioengineering 2023 (CC BY). Lock delays: MRBA 0.28 s, IMU walker 0.8 s.

What is new: CatchPoint vs existing systems

	Mobile overground	Predictive trigger XCoM + delay compensation	Patient-specific step reach	Graded minimum force	Preserves self-recovery practice	Catches only when needed	Nudge + catch in one cable path	Passive fail-safe force limit
NTU MRBA	●	○	○	○	○	○	○	?
NTU DRBA	?	○	○	○	○	○	○	?
Hocoma Andago	?	○	?	○	?	○	?	?
KineAssist	●	○	○	○	◑	○	○	?
RobUST (Columbia)	○	○	◑	?	●	○	◑	?
mTPAD (Columbia)	●	○	○	◑	?	○	○	?
ZeroG	◑	○	○	○	?	?	○	?
NIMBLE (URJC)	●	?	?	?	?	?	○	?
ExoRecovery (exo)	?	●	?	○	○	?	○	?
CatchPoint (ours)	◑	●	●	●	●	●	◑	◑

● yes ◑ partial ○ no ? not reported in sources we read

Based on published descriptions we could access; ? = not reported. CatchPoint: simulation + design only.

To our knowledge, none of the systems we read combines a predictive, per-patient trigger with a graded minimum force and a catch only when needed. Cells marked ? are not reported in the sources we read; we do not count them against a system.

Source: team comparison of published system descriptions we could access (per-cell sources in the notes). CatchPoint: simulation and design only.

Three contributions, each against its closest prior work

New control approach

Recoverability-based trigger

Act when the predicted capture point leaves this patient's own step reach.

Closest prior work
vs MRBA / DRBA / RobUST: fixed threshold or static boundary

Evidence: simulation + physics demo

New control algorithm

Graded minimum nudge

Smallest pelvic force (≤ 150 N) that restores a reachable step; catch only if not enough.

Closest prior work
vs threshold-lock walkers (on/off) and always-on support

Evidence: simulation + physics demo

New interface / actuation (design)

One cable set, three modes

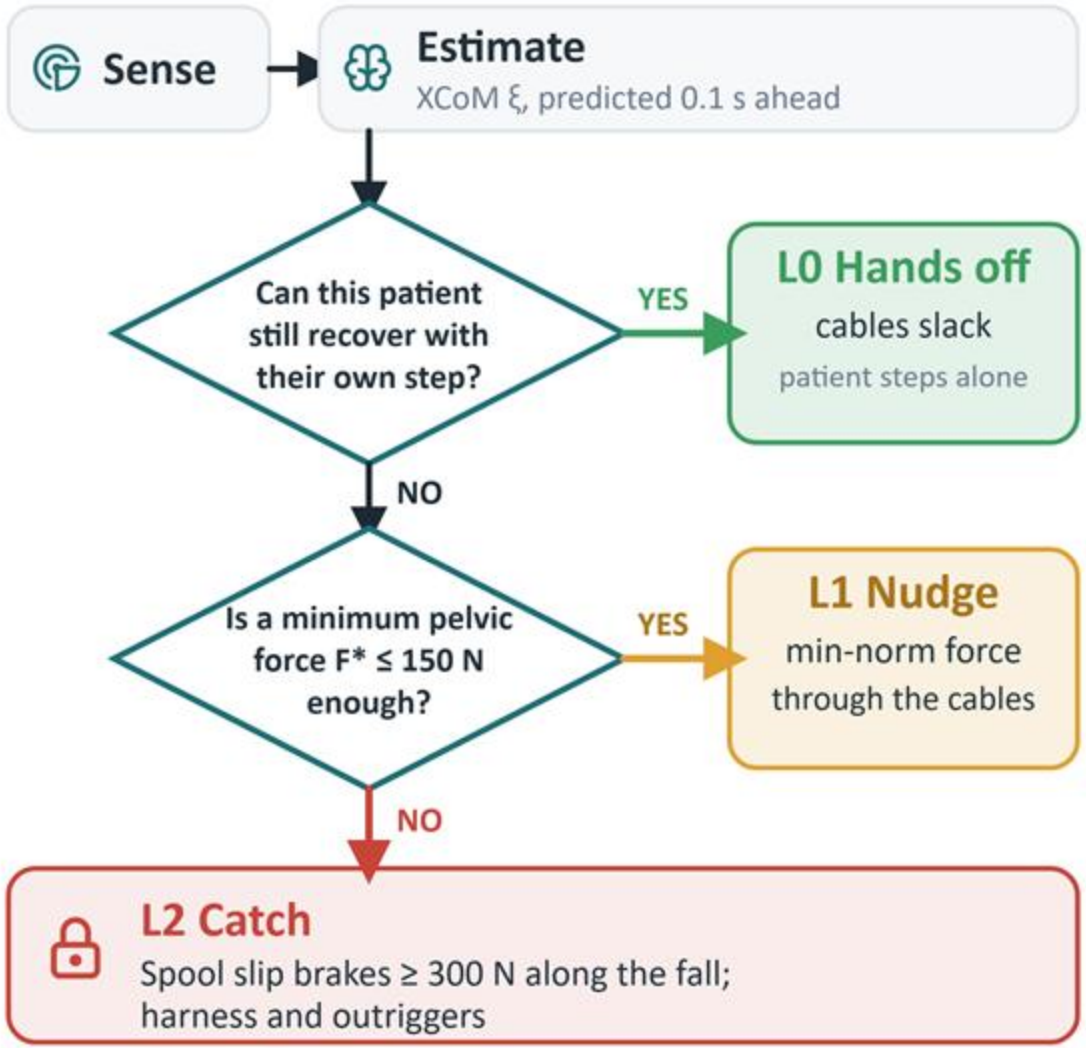
Servo nudges and power-off slip brakes catch, on the same cables.

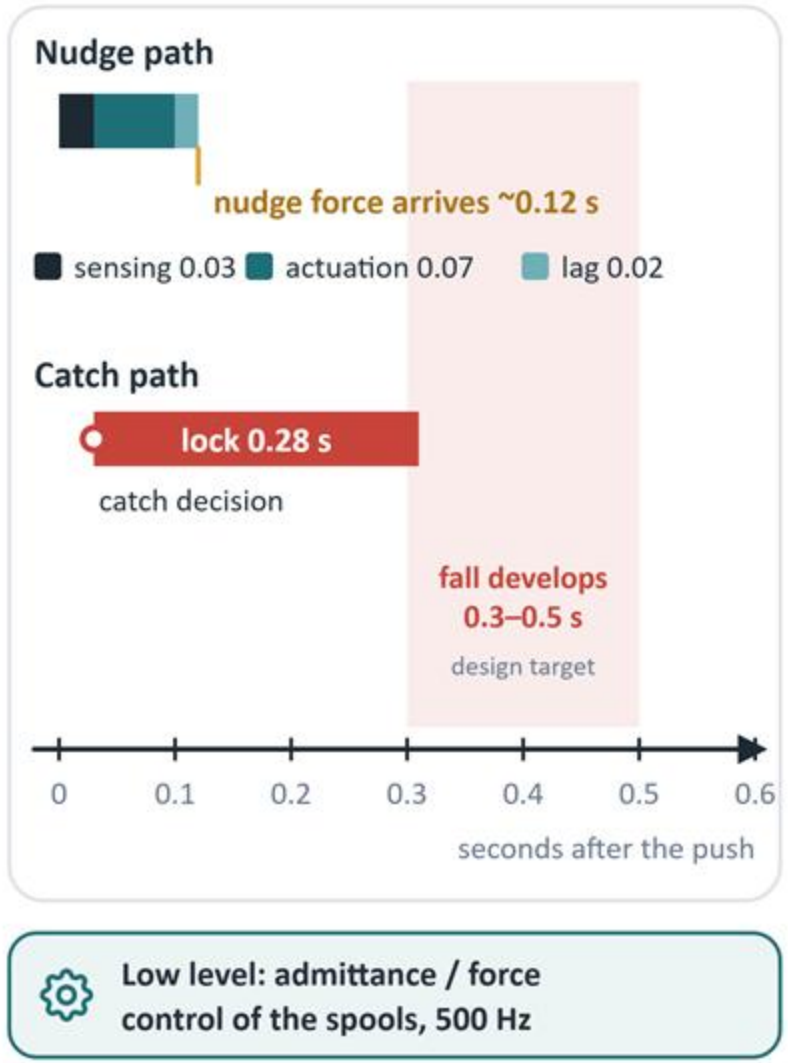
Closest prior work
vs mTPAD (no catch); NIMBLE, ZeroG (catch on a separate or overhead path)

Evidence: design analysis; unbuilt

Categories follow the course briefing. Physics demo: calibrated on the same push grid; robot-policy demonstration only. Novelty "to our knowledge" (search-based, not proof).

Control architecture and timing





Two questions, in order

Recover alone? If not, is ≤ 150 N enough?

After a catch lock

COM stops 0.10 s later (median).

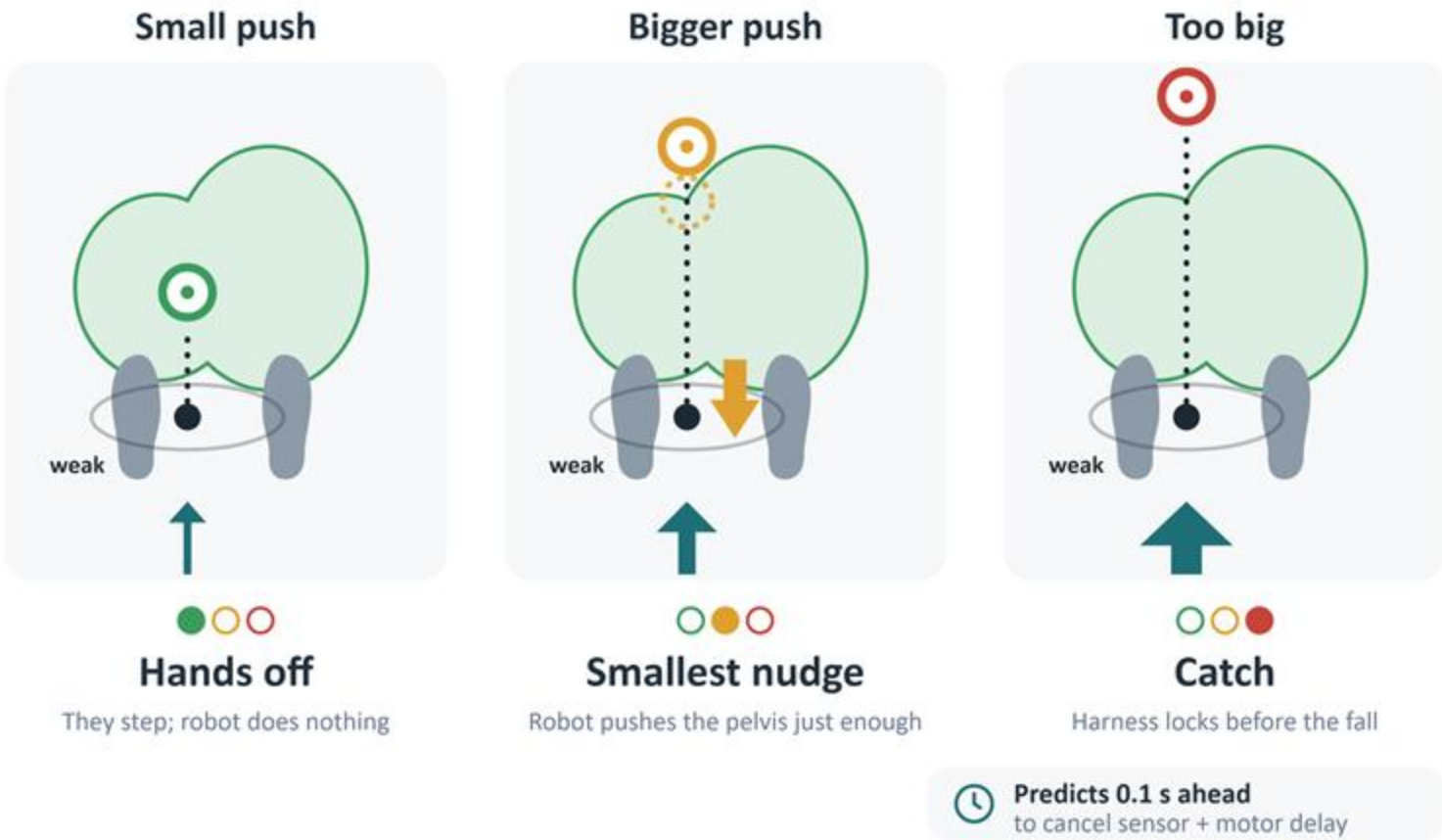
All delays

are model assumptions.

Figure by the project team. Delays (sensing 0.03 s, actuation 0.07 s, lag 0.02 s) are model assumptions, not measurements; COM stop times from the team's sizing simulation (passive slip brakes, lock 0.28 s).

Where they must step vs. where they can reach

Ask "can they still save themselves?" — not "have they started to fall?"



How to read it

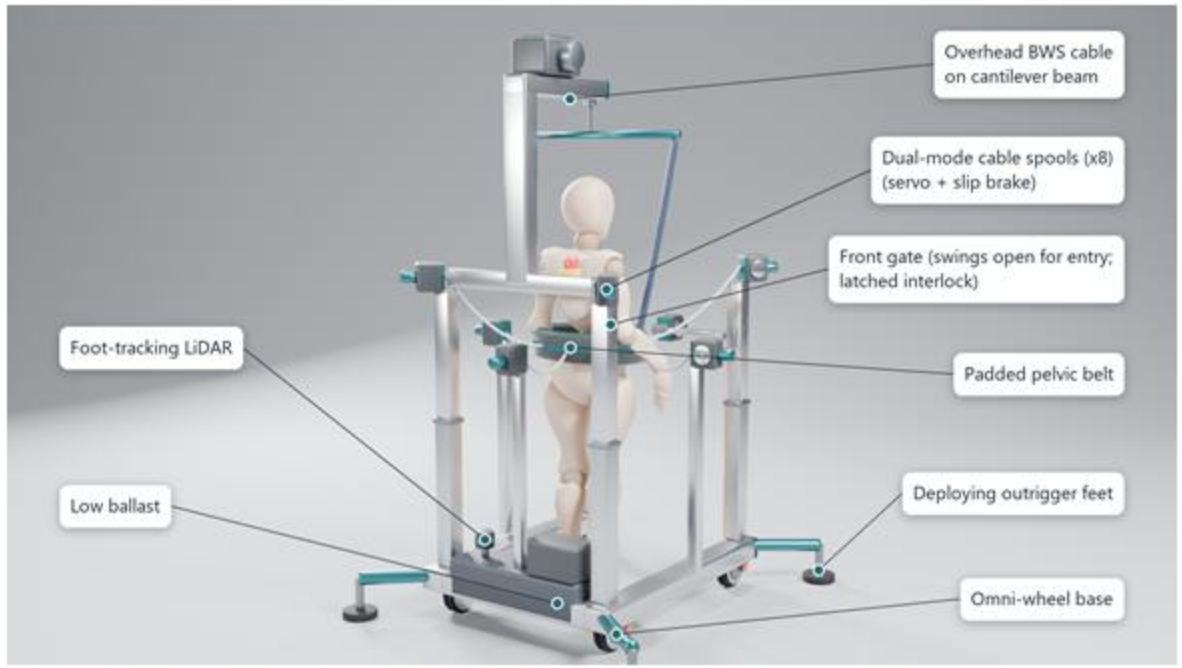
◎ = where they must step to stay upright (capture point)

Green area = where this patient's own step can reach

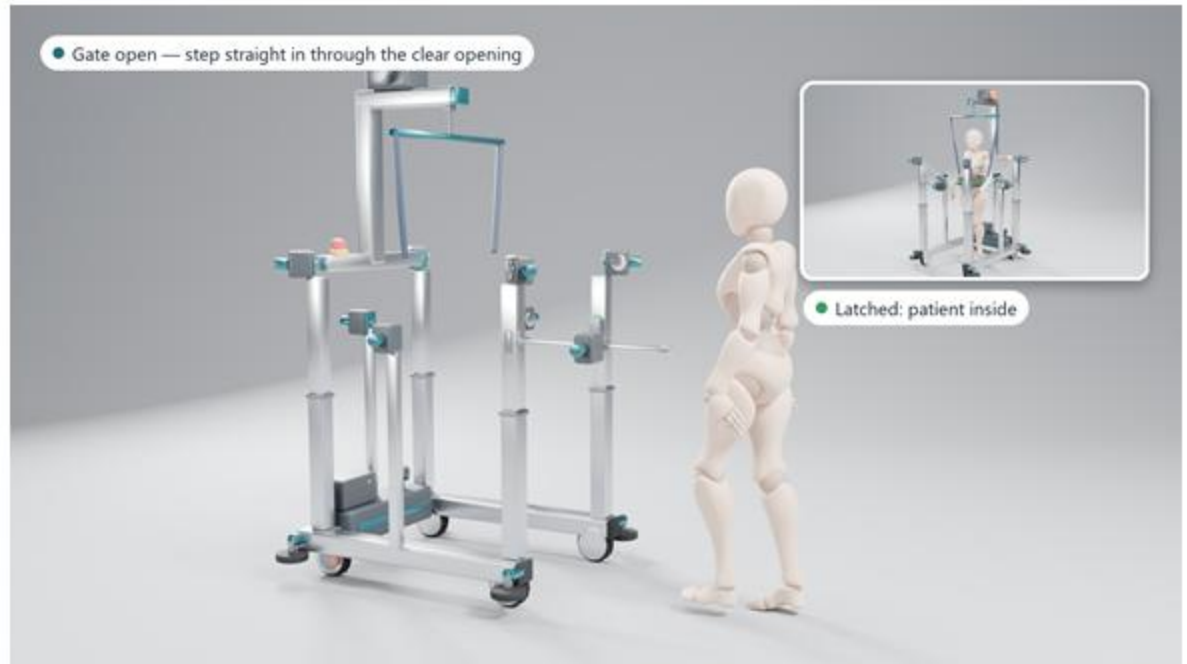
Inside it: hands off. Outside: smallest nudge. Out of reach: catch.

Schematic illustration, top view, by the project team. The capture point is a standard balance concept (Pratt 2006; Hof 2005).

Hardware: one cable set, three modes



Design D3: cables, dual-mode spools, overhead body-weight cable, outriggers



Latched front gate: opens to step in, latched during use

One cable set

Slack · nudge ≤ 150 N · catch ≥ 300 N

Dual-mode spools

Servo nudges; power-off slip brake catches

Designed not to tip

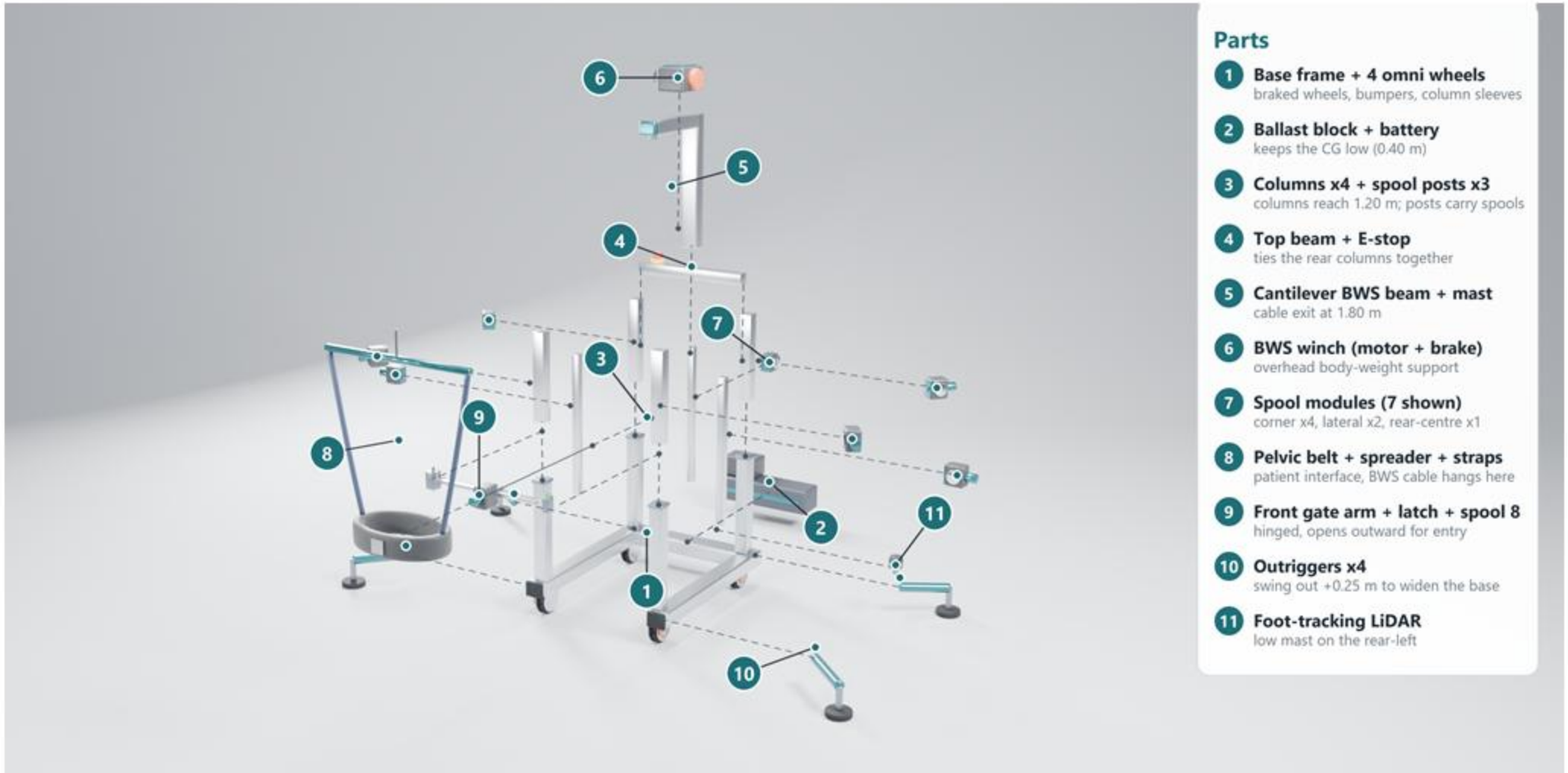
Tip SF ≥ 1.5 up to 420 N (statics)

Concept, unbuilt

Statics and simulation only

Render: Blender (design D3 CAD). Mannequin: Mixamo/Adobe via three.js examples; HDRI/texture: Poly Haven (CC0).

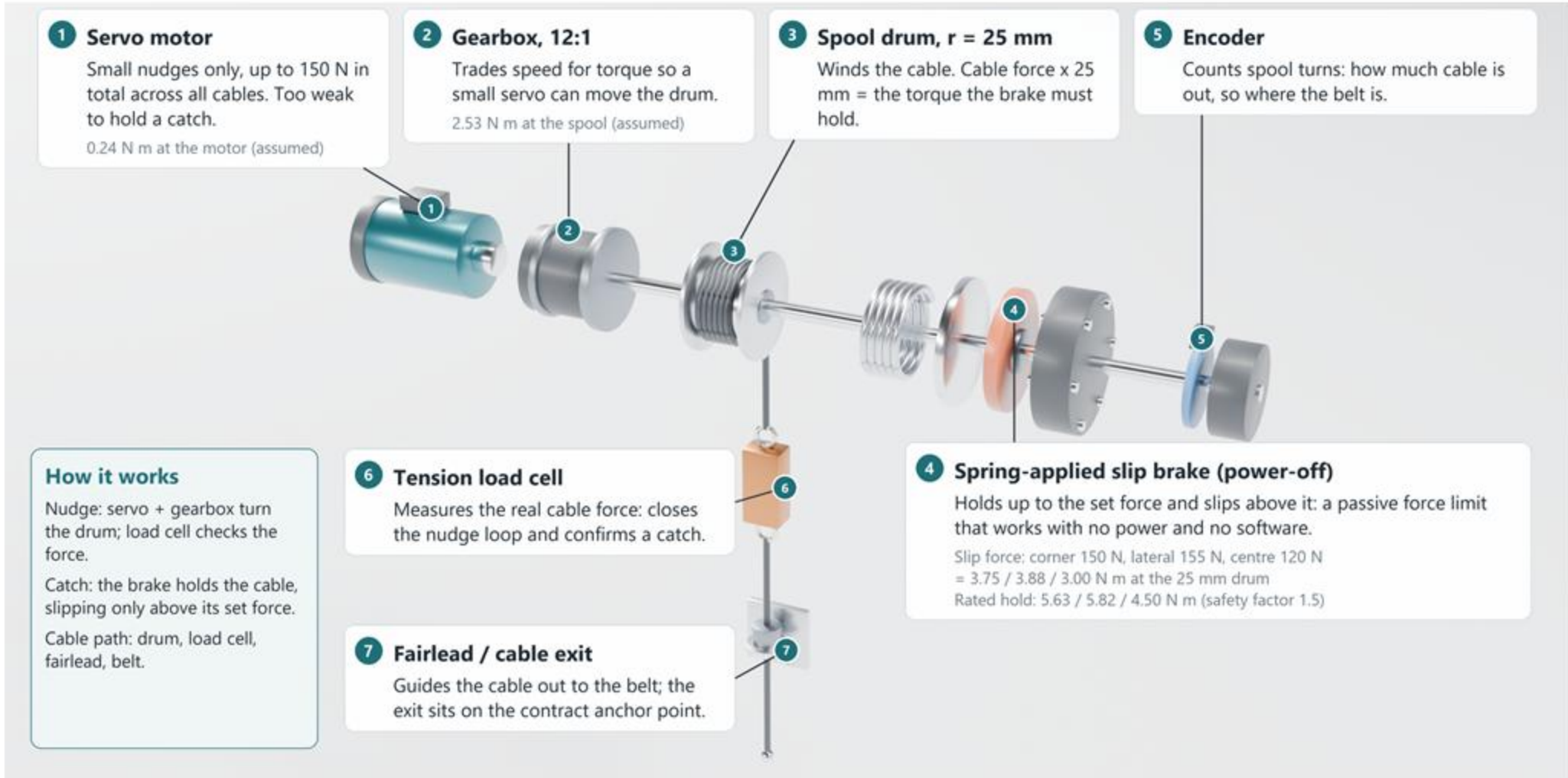
Design D3: exploded assembly



Parts pulled apart along their mounting axes; dashed lines show where each part mounts. Cables omitted for clarity. Concept render: unbuilt and unvalidated.

Render: Blender (design D3 CAD).

One dual-mode spool: servo for nudges, slip brake for the catch

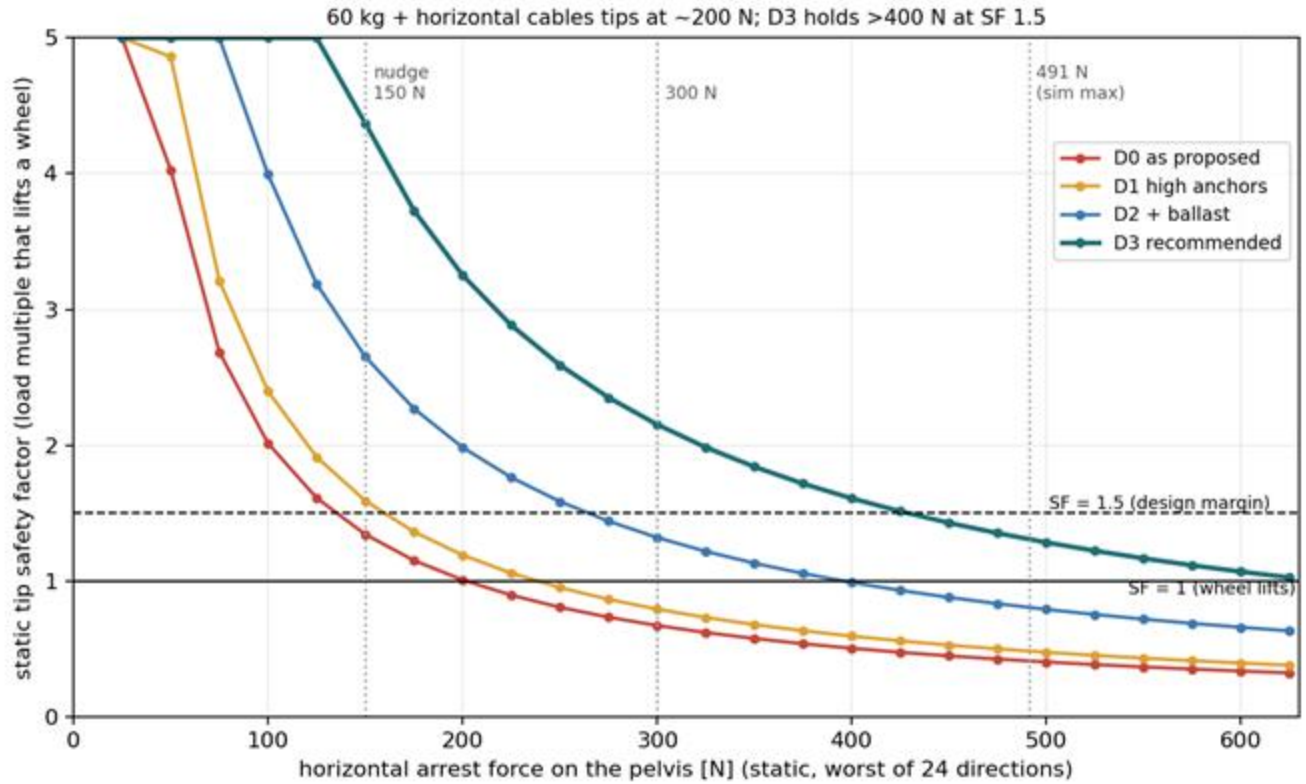


8 identical modules per walker; part shapes are placeholders, not a part selection. Concept render: unbuilt and unvalidated.

Render: Blender (design D3 CAD).

Can a walker catch without tipping?

Statics + simulation: an 80 kg low-CG walker with outriggers is not predicted to tip.



805/814

caught trials held by the passive slip brakes (98.8%, CI 97.9–99.5%)

377/814

same trials, original 60 kg design (46.3%)

380 ≤ 420 N

max slip-brake force vs tip limit (SF 1.5)

Not validated. The other 1186/2000 trials never need the catch, so this is not a fall-reduction claim.

Source: team statics analysis of design D3 and one seeded 2,000-trial simulation (CatchPoint, lock 0.28 s). Static tip safety factor, worst of 24 directions.

Same push, three responses



Hands off: cables slack

Nudge: ≤ 150 N

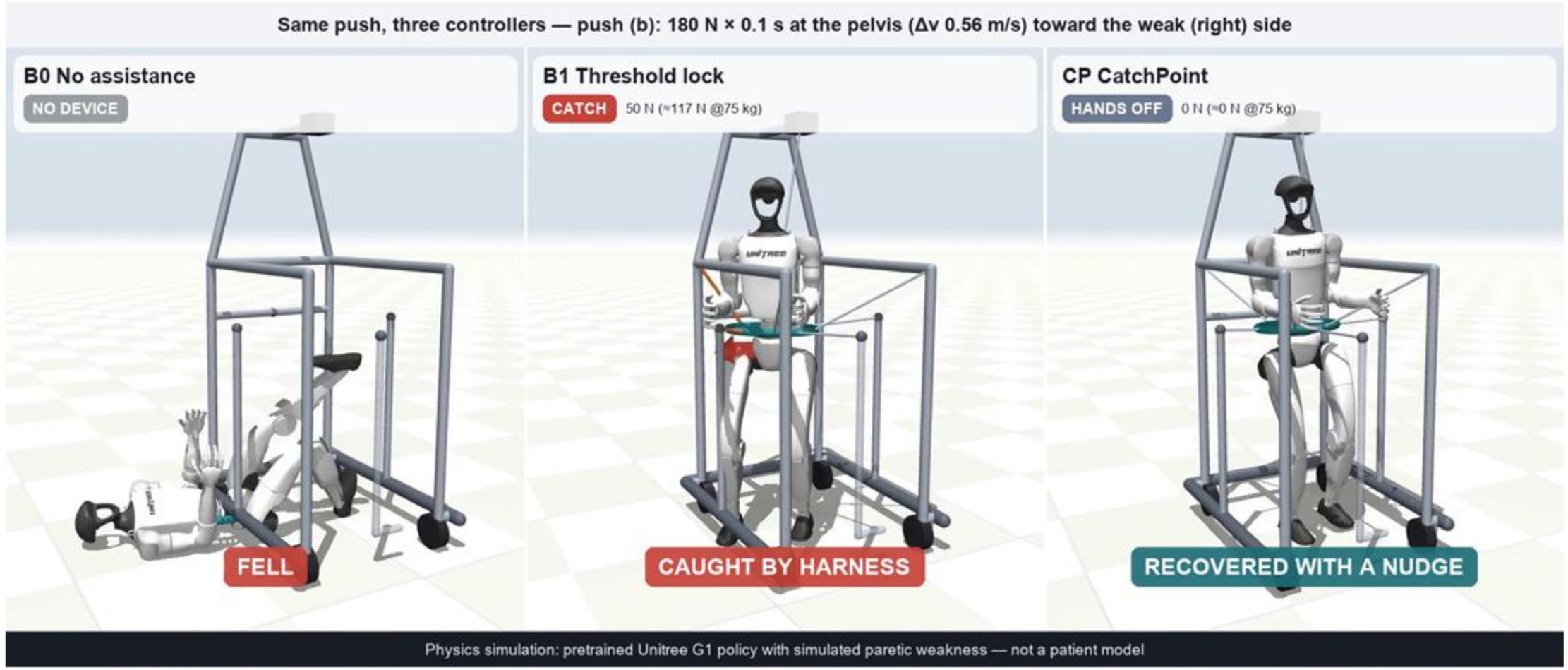
Catch: slip brakes ≥ 300 N along the fall, ≤ 380 N total

One cable set, three modes: slack · graded nudge ≤ 150 N · passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total

Design limits. Concept render of design D3: unbuilt and unvalidated.

Render: Blender; mannequin Mixamo/Adobe via three.js examples; HDRI/texture Poly Haven (CC0).

The same push in physics simulation: fall, lock, or nudge

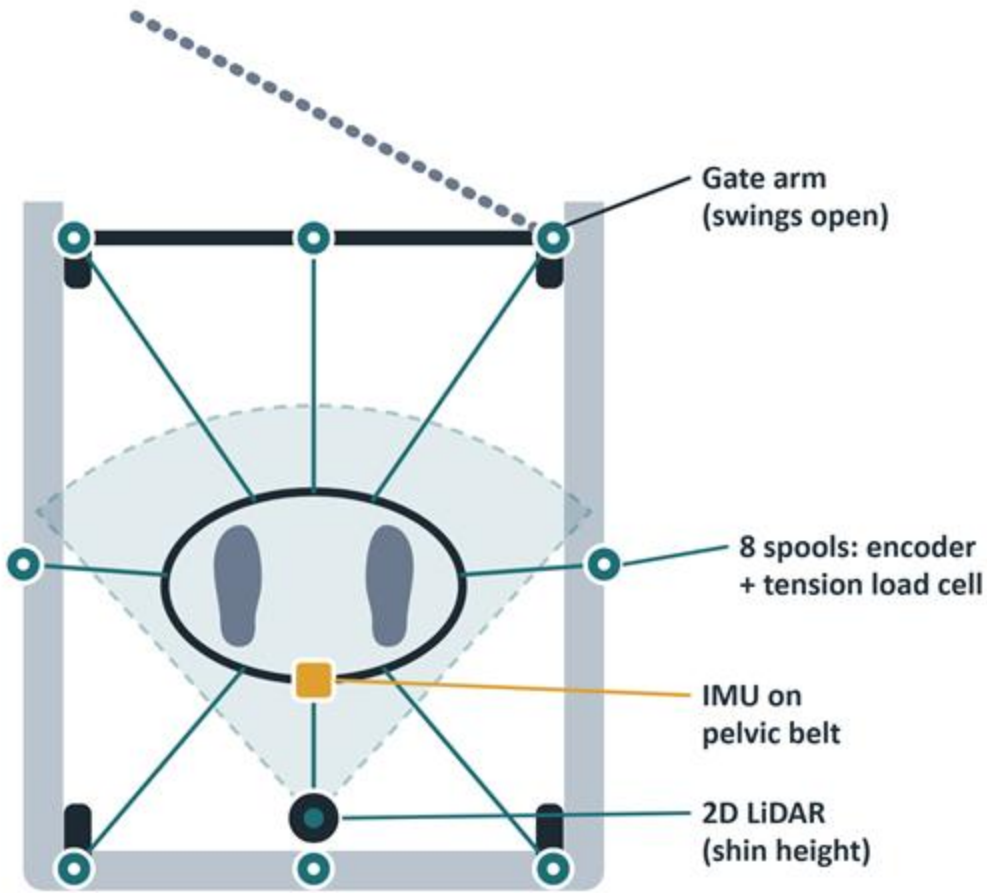


180 N push toward the weak side. No device: falls · threshold lock: harness catch · CatchPoint: a small nudge.

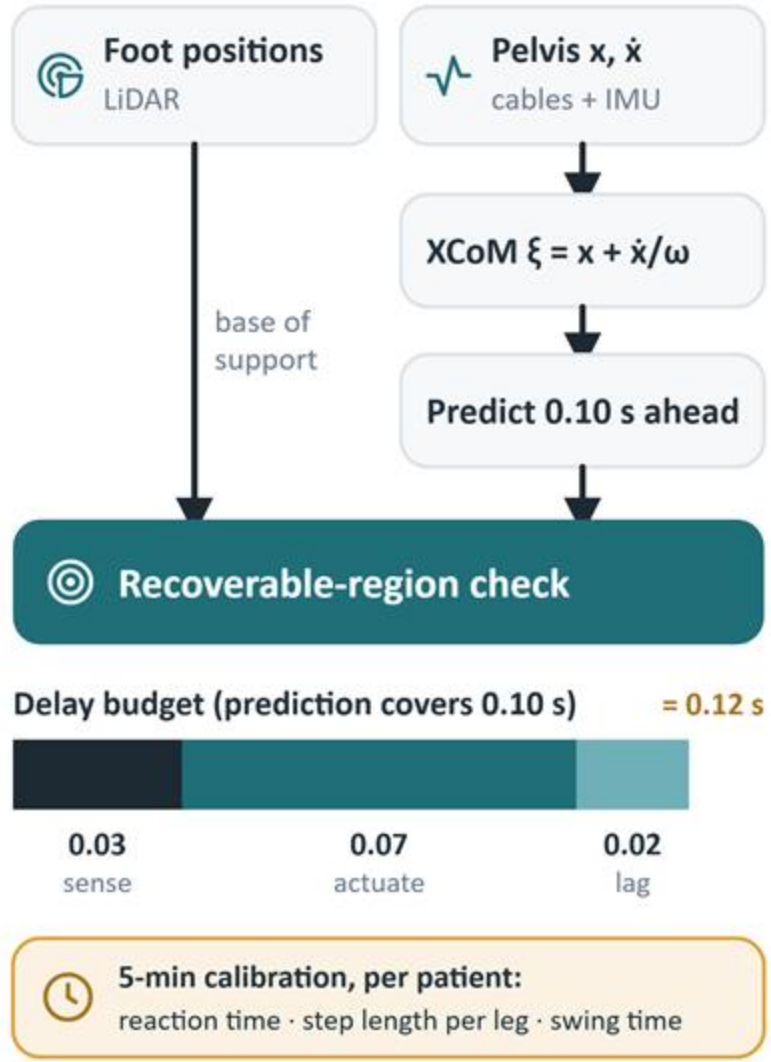
Robot-policy demonstration (Unitree G1 in MuJoCo), not a patient. Click the picture to play.

Source: MuJoCo physics demo, Unitree G1 policy with simulated weak right leg (one deterministic run).

Sensing and state estimation



D3 walker, top view · 0.95 × 0.80 m



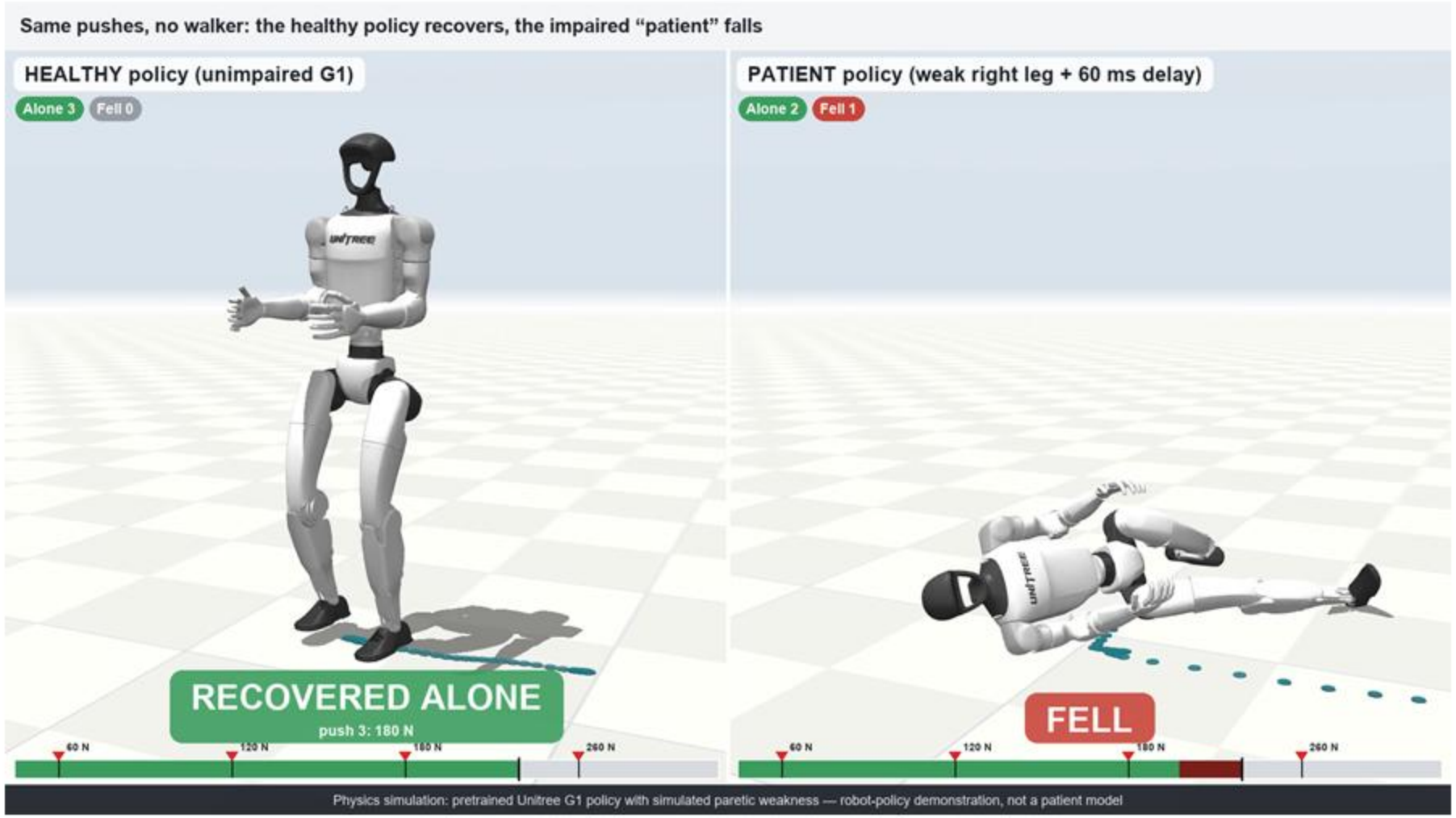
Sensed: feet, pelvis, cable force

Estimated: XCoM, 0.10 s ahead

Calibrated: 5 min per patient

Figure by the project team. Sensor choices are proposals, not tested hardware.

The physics patient: same pushes, healthy recovers, impaired falls



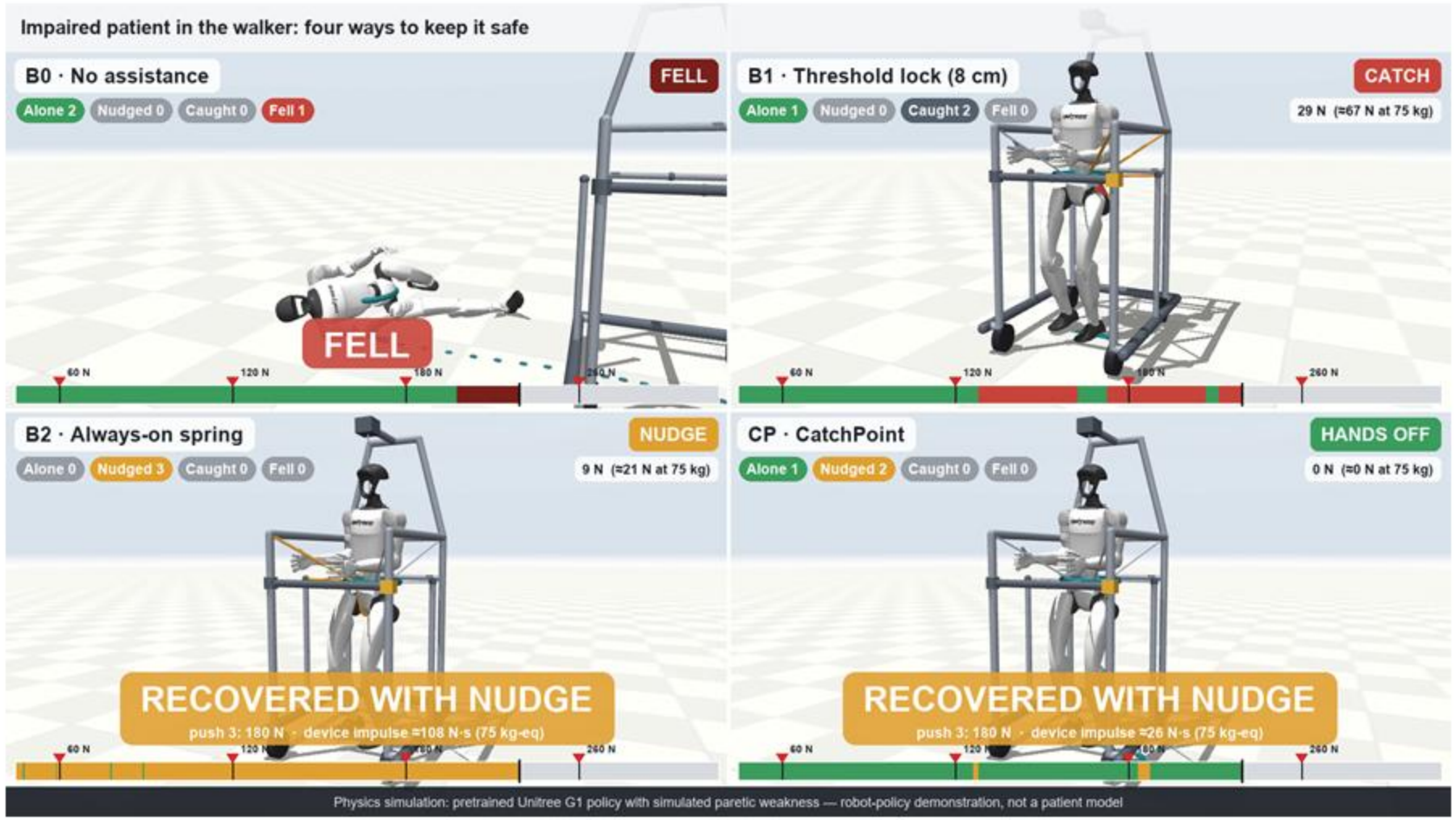
Pretrained Unitree G1 walking policy in MuJoCo.

Patient: right-leg torque ×0.5 plus 60 ms delay.

Four weak-side pushes (60 / 120 / 180 / 260 N): the healthy policy recovers from all four; the patient falls at 180 N.

Source: MuJoCo physics demo, Unitree G1 policy (one deterministic run). Robot-policy demonstration only — not a patient model. Click the picture to play.

Physics demo: fewer catch-locks, more low-force nudges

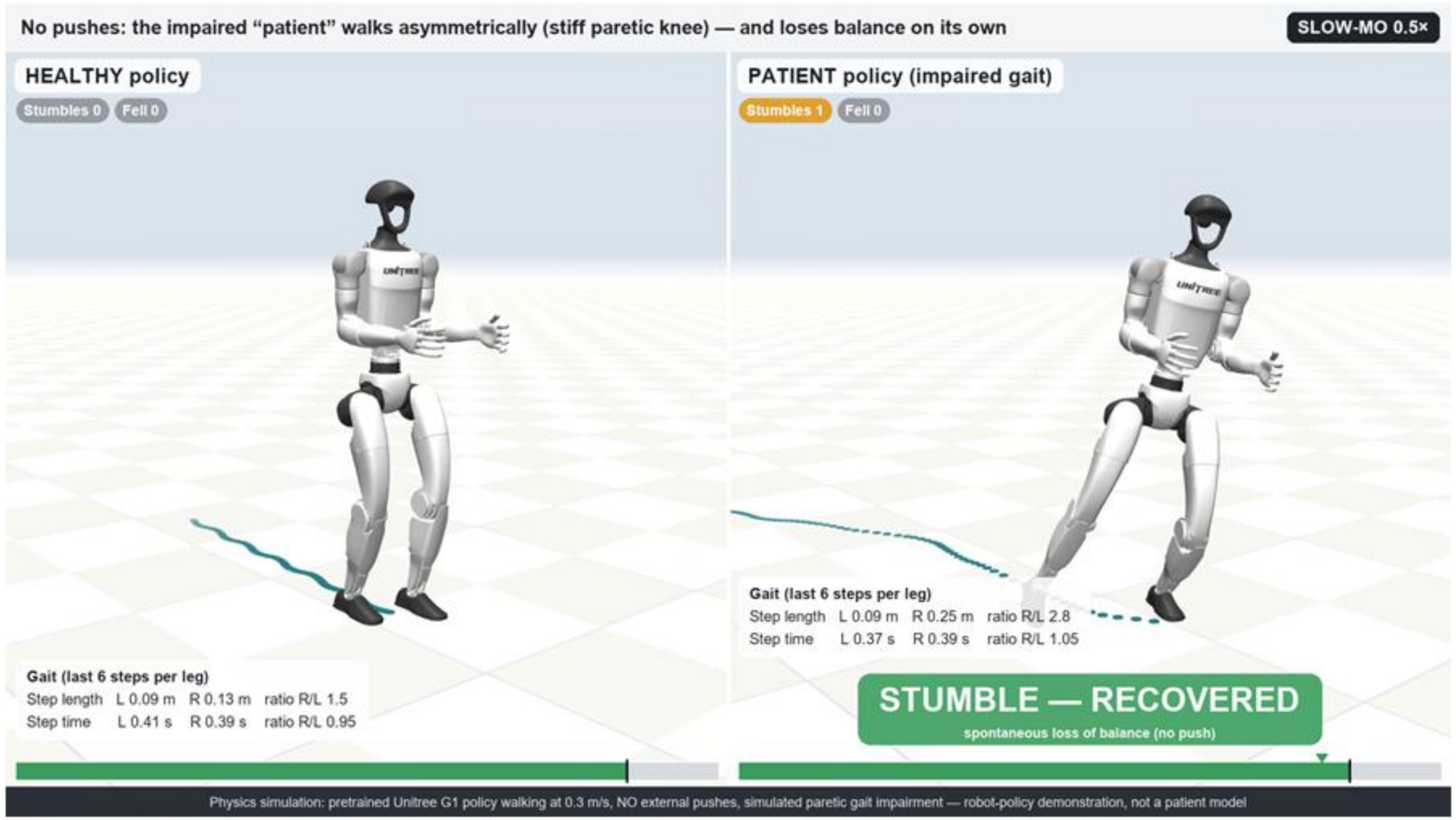


Same weak-side pushes, four ways: no robot, threshold lock, capture-point threshold, CatchPoint.

Over the 56-push grid CatchPoint: 0 falls, 26 catch-locks, 23 nudges.

Source: MuJoCo physics demo, Unitree G1 policy (BSD-3). Robot-policy demonstration only — not patient evidence. Click the picture to play.

No push needed: the impaired gait shows on its own



Robot-policy demonstration only — not patient evidence. Click the picture to play.

Source: MuJoCo physics demo, Unitree G1 policy (walking, v3 protocol).

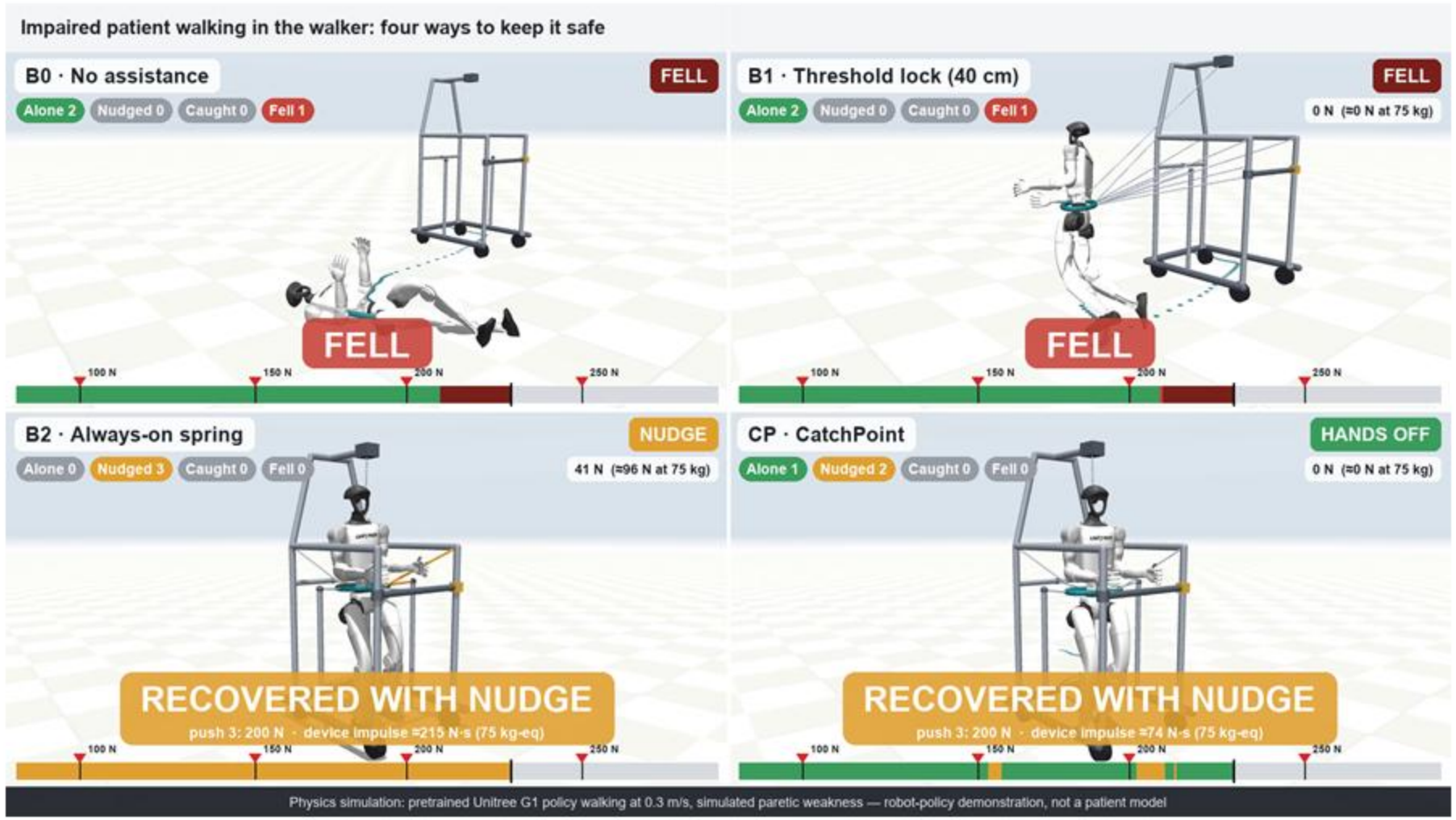
Walking at 0.3 m/s, no pushes.

Patient: right-leg torque ×0.5, 60 ms delay, stiff paretic knee.

Paretic knee bends 36° vs 46° healthy.

Falls on its own in 13/20 walks; healthy 0/20.

Walking in the walker: four ways to keep it safe



Robot-policy demonstration only — not patient evidence. Click the picture to play.

Source: MuJoCo physics demo, Unitree G1 policy (walking, v3 protocol).

Walking patient, same pushes: no robot, threshold lock, always-on spring, CatchPoint.

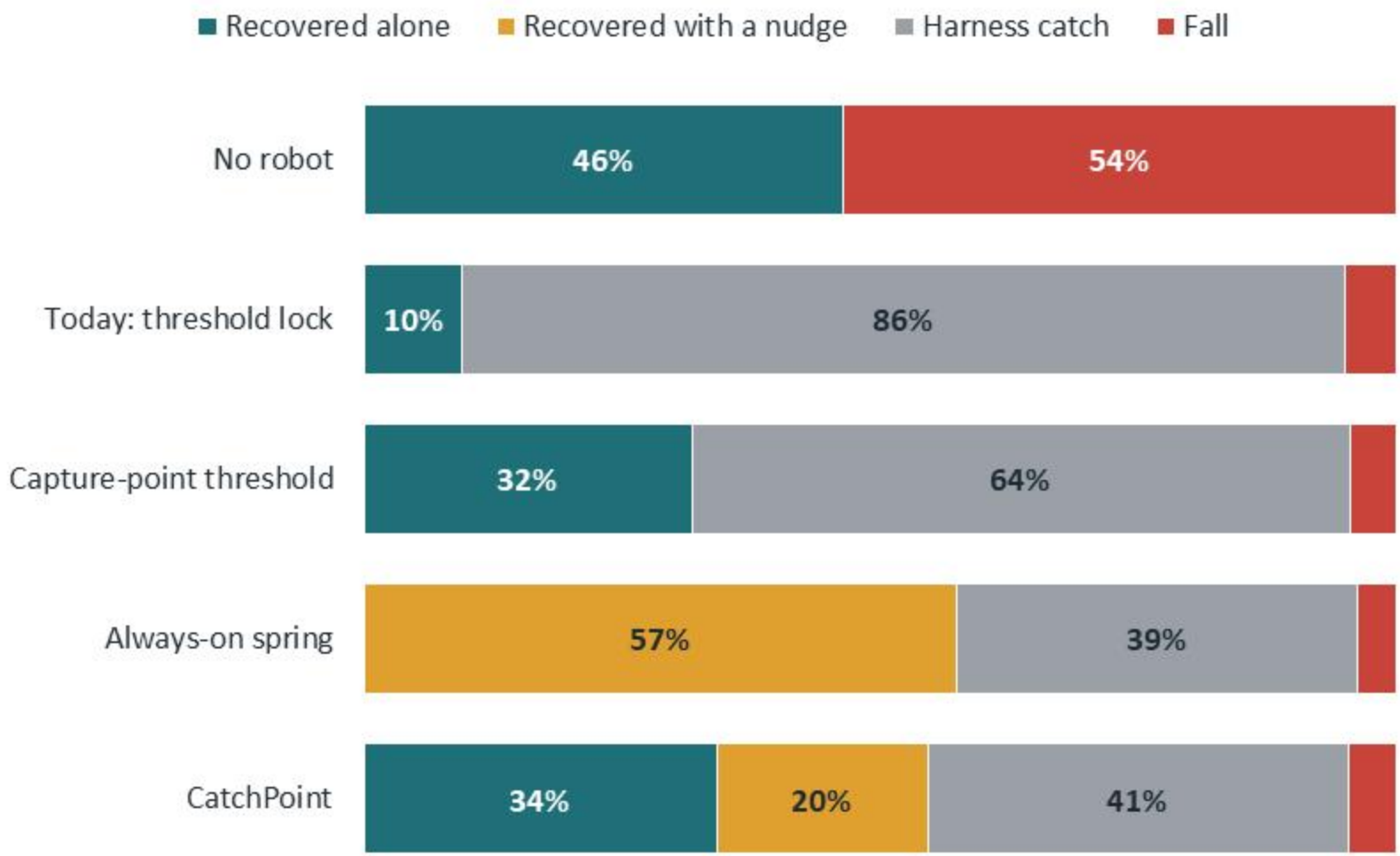
Held-out test, 48 pushes

CatchPoint: 0 falls, 7 locks

Threshold lock: 5 falls, 12 locks

No robot: 17 falls

At similar fall rates, far fewer harness catches



Ended without the harness

54.6% vs 9.6%

34.2% alone + 20.4% nudged, vs 9.6% for threshold lock; both ~5% falls.

Capture-point threshold: harness 63.7% vs 40.7%. CIs ≤ ±4 pp; simplified model.

[Try it: interactive demo ↗]

Source: team Monte Carlo simulation, 2,000 trials per controller, seed 5412. Selected operating points: threshold lock 0.05 m, capture-point threshold 0.08 m, spring 200 N/m, CatchPoint margin 0.03.

Limitations and next steps

What we have not shown

1 Simplified balance model; patient parameters assumed

2 Physics demo: robot policy; CatchPoint still fell in 9/20 no-push walks (trips)

3 Hardware unbuilt: statics only; outriggers and spool parts not selected

4 No patient data: all results are simulation

Next steps

1 Bench-test one dual-mode spool

2 Handle forward trips; test other speeds and impairments

3 Clinician feedback on thresholds and stop rules

4 Pilot study (next slide) after ethics approval

Closing the loop on the claims made in this talk.

Clinical evaluation plan: proposed, not conducted

Proposed protocol — not conducted

Participants

Chronic stroke

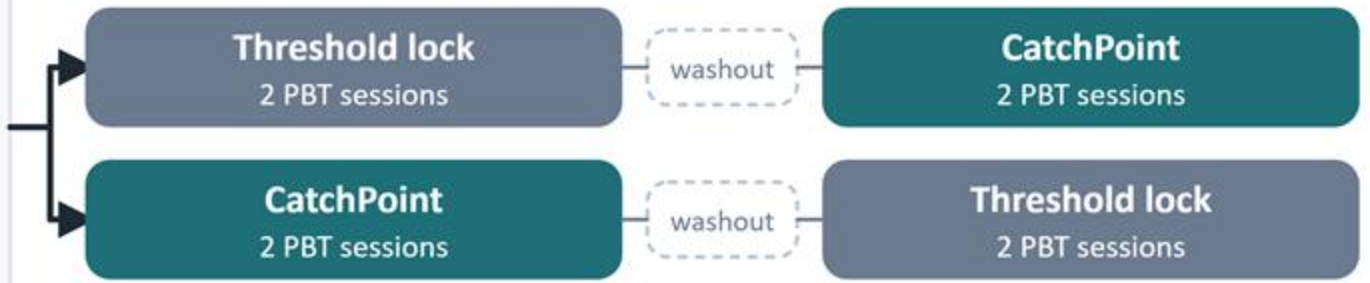
Stands ≥ 30 s unaided

Able to step

n = 10–12

n: design target

Randomised within-subject crossover



Order randomised; each person does both

Every session

- Falls (target 0)
- Harness catches
- Self-initiated recovery steps
- Assisted recoveries
- Robot impulse

Before / after each period

- Berg Balance Scale
- Mini-BESTest
- Reactive step latency and length
- FES-I (fear of falling)

Safety

- Stop: any injury
- Stop dose: ≥ 2 unplanned catches in a row
- Therapist spotter
- Ethics approval (IRB)

Targets (n, falls, stop rule) are design proposals

Question

Does CatchPoint keep more self-initiated recovery steps than threshold lock, without more falls or catches?

Therapist spotter, stop rules and ethics approval come first.

Figure by the project team. Sample size, fall target and stop rule are design proposals. No participants have been recruited.

Summary

1

A new trigger

Act when recovery is no longer feasible for this patient — not when a fixed threshold trips.

2

A graded response

Hands off, minimum nudge, catch only when needed; one cable set is designed to do all three.

3

Evidence so far

In simulation: harness in 40.7% of pushes vs 85.5% for threshold lock at similar falls. Design analysis only for the hardware.

A Clinical · B Mechanical · C Sensing · D Control · E Simulation

Public page: https://chuanhan1013.github.io/catchpoint/

Interactive simulator demo ↗

Appendix

Technical depth for anyone who wants it

Control law · Simulation model · Results · Robustness · Design D3 analysis · Prior work · Physics-demo protocol · References

A1 · Control law: predict, check, push no more than needed

Where must the foot land? (capture point)

$$\xi = x + \dot{x} / \omega$$

$\omega = \sqrt{(g / z_0)}$; x, $\dot{x}$ = centre-of-mass position and velocity.

Where will it end up with push F?

$$\xi_{land}(F) = p + (\xi - p)\, e^{\omega T} + \beta F$$

ξ = predicted capture point; p = centre of pressure; T = time to foot landing; $\beta = (e^{\wedge}\omega T - 1)/(m\omega^2)$.

Smallest push that keeps the step reachable

$$F^* = \operatorname{argmin} |F|^2 \quad \text{s.t.} \quad h(F) = r - |\xi_{land}(F) - a| \geq 0$$

r = the patient's step reach minus a margin; a = stance foot. Closed-form solution.

Which level?

- h(0) ≥ 0: patient recovers alone, F = 0
- |F*| ≤ 150 N: command F*
- |F*| > 150 N: command 150 N and catch

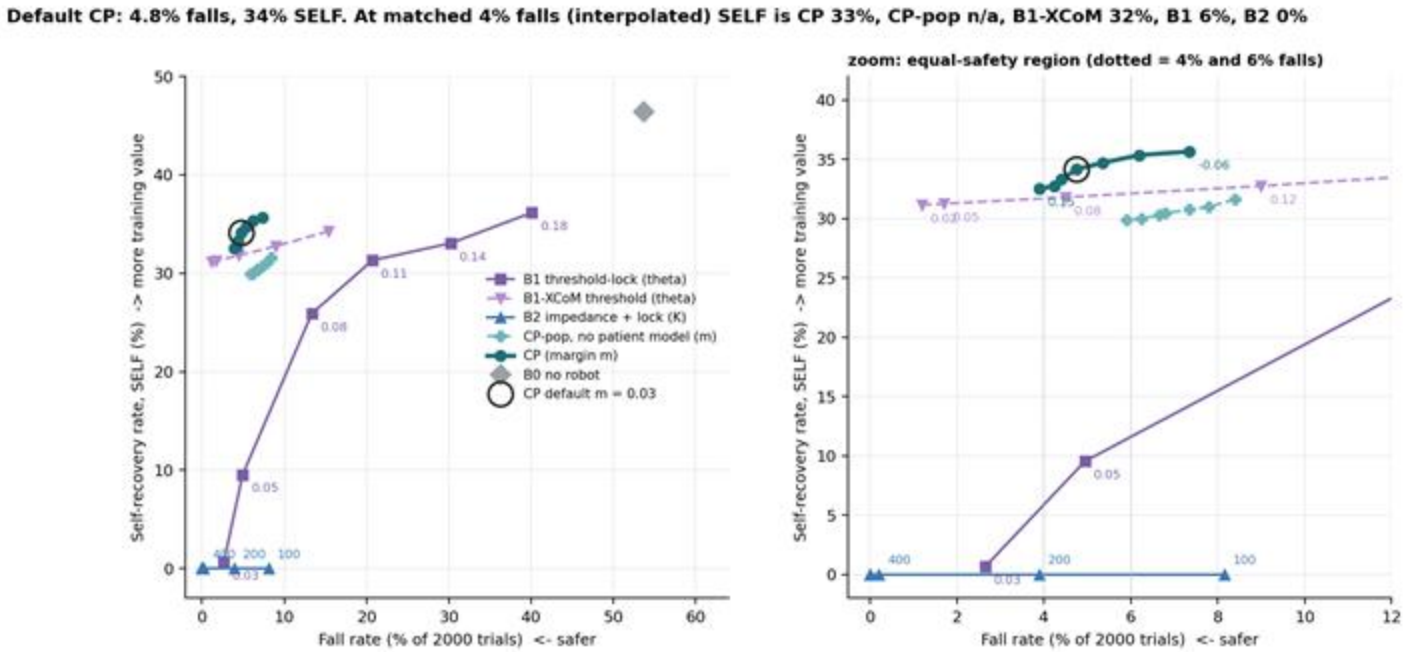
Sources: Hof et al., J Biomech 2005; Pratt et al., Humanoids 2006. Control law: our own design.

A2 · Simulation model and virtual patients

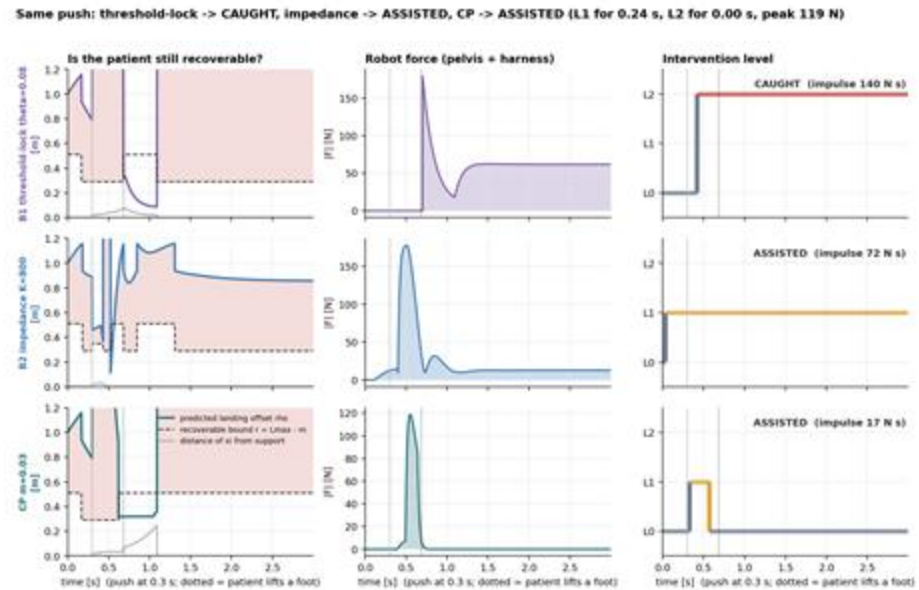
Parameter	Value or range	Note
Balance model	3D linear inverted pendulum	Horizontal plane, 2 ms step
Trial	3.0 s; push at 0.3 s	Δv 0.2–0.9 m/s, any direction
Population	50 patients × 40 pushes	2000 trials; same set for every controller (seed 5412)
Severity s	uniform 0–1	Drives all patient limits below
Mass / COM height	60–85 kg / 0.85–1.00 m	Sampled per patient
Reaction time	0.20 + 0.30·s	Seconds
Max step (paretic / other)	0.50 − 0.30·s / 0.60 − 0.10·s	Metres
Swing time (paretic / other)	0.40 + 0.25·s / 0.35 + 0.10·s	Seconds
Robot delays	sensing 30 ms; actuation 70 ms + 20 ms lag	Catch lock delay 0.28 s
Limits	Assist ≤ 150 N; fall = COM 0.25 m beyond support	Catch: harness spring-damper

All ranges are modelling assumptions anchored to literature, not clinical data. s = severity.

A3 · More practice kept at the same fall rate



Each controller traces a trade-off between falls and self-recovery as its setting is swept.



One showcase push: CatchPoint waits, nudges, and escalates only when needed.

Source: team Monte Carlo simulation, seed 5412.

A4 · Fair comparison at matched fall rates, with confidence intervals

Solo recoveries, interpolated at equal falls

Controller	Solo recoveries at 4% falls	at 6% falls
Threshold lock (B1)	5.9%	11.6%
Capture-point threshold	31.7%	32.1%
Always-on spring (B2)	0.0%	0.0%
CatchPoint (CP)	32.6%	35.2%
CP, population model	n/a (not bracketed)	29.9%

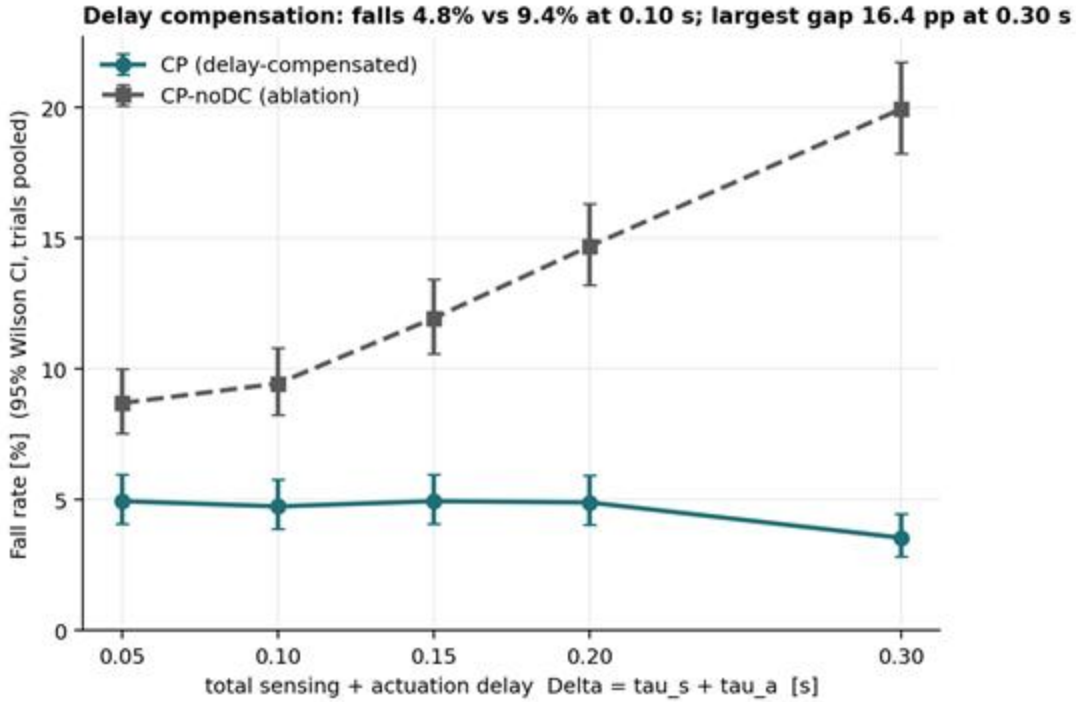
95% patient-bootstrap confidence intervals

Setting	Falls [95% CI]	Solo [95% CI]
Threshold lock 0.05	5.0% [3.4–6.6]	9.6% [7.2–11.9]
Spring 200 N/m	3.9% [2.9–4.9]	0.0% [0.0–0.0]
CatchPoint 0.03	4.8% [3.4–6.2]	34.1% [31.3–37.0]
CatchPoint 0.15	3.9% [2.7–5.3]	32.5% [29.8–35.0]

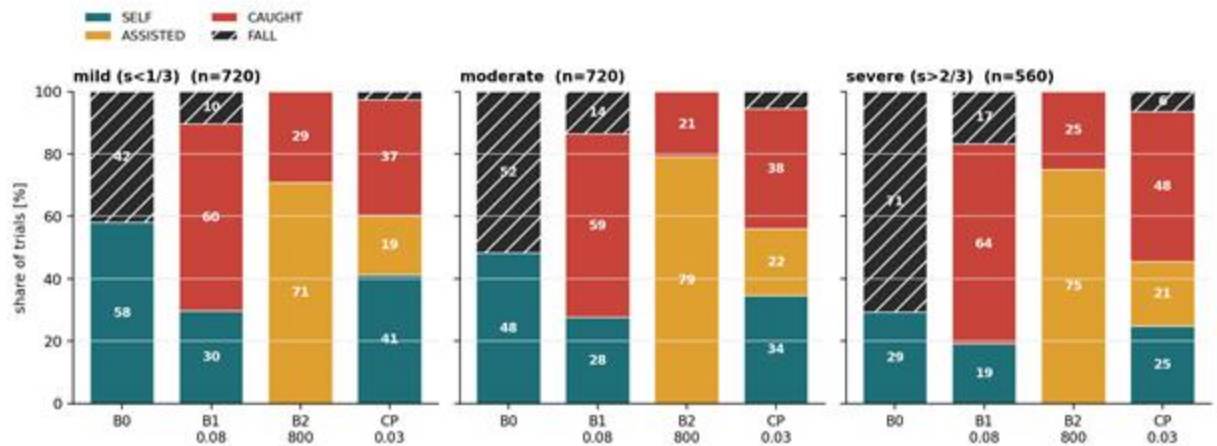
Ablation: the patient-specific model (CP) versus a population-average one (CP-pop, margin 0.03) is worth about 2 pp fewer falls and 4 pp more solo recoveries.

Source: team Monte Carlo simulation, seed 5412. Interpolated between swept settings.

A5 · Robustness: sensing delay and patient severity



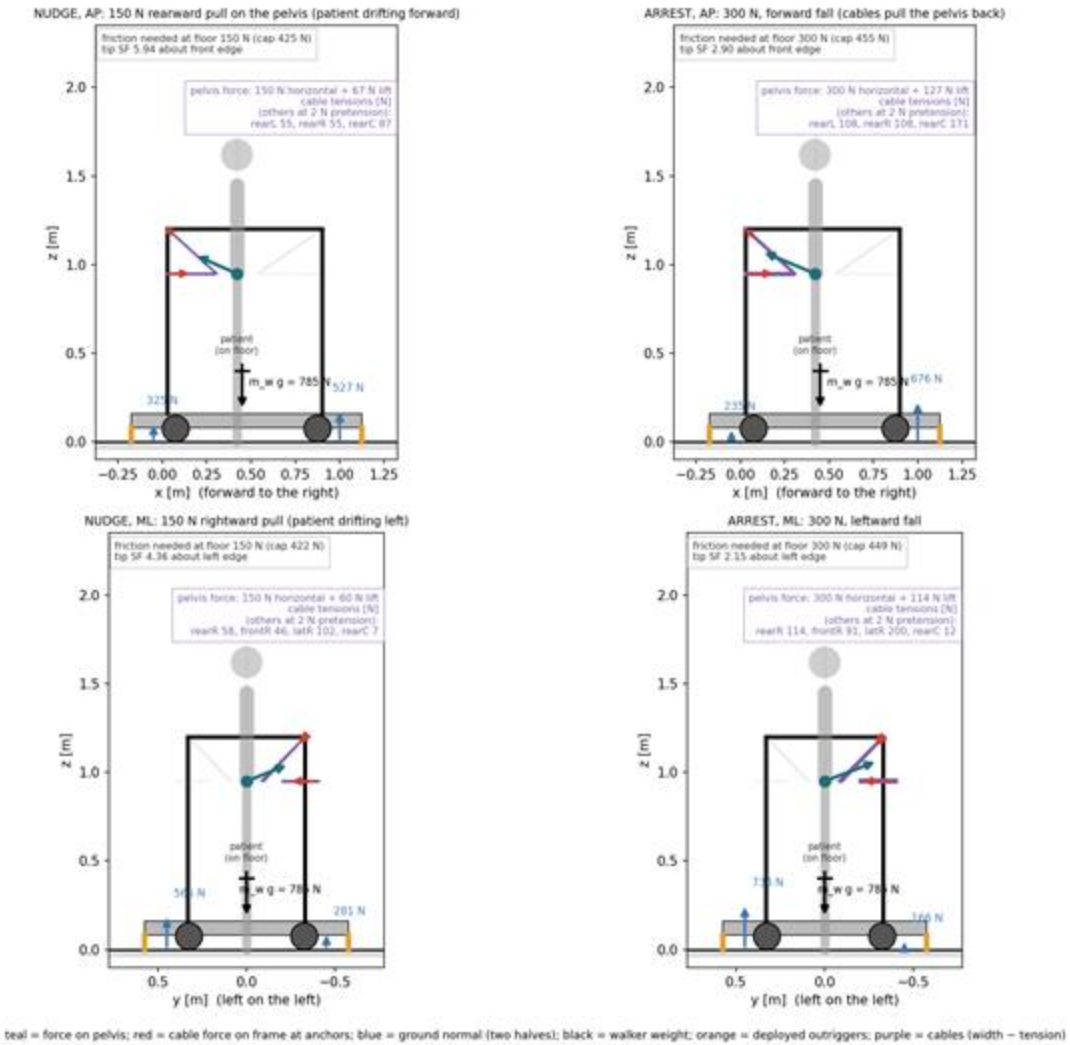
Without delay compensation falls rise with latency; with it they stay flat.



Outcomes by patient severity (B0 none, B1 threshold lock 0.08, B2 spring 800, CP = CatchPoint margin 0.03). In the severe group CatchPoint falls 6% and keeps 25% self-recovery.

Source: team Monte Carlo simulation, seed 5412.

A6 · Free-body diagrams: nudge and arrest



Nudge, 150 N: tip safety factor 5.94 (AP) and 4.36 (ML); lift on the pelvis ≤ 66.1 N.

Arrest, 300 N, controller-set tensions: tip safety factor 2.90 (AP) and 2.15 (ML); friction needed 300 N against a cap of 455 N (AP).

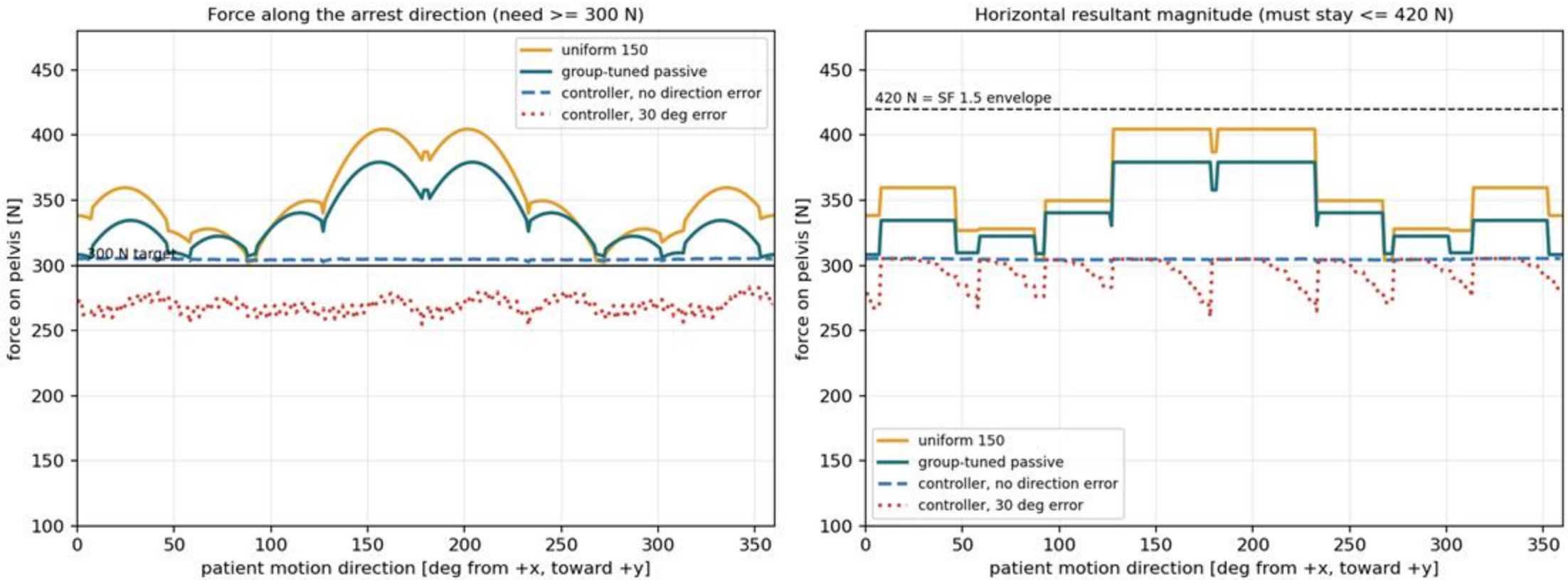
Passive slip brakes (next slide): ≥ 300 N along the fall, ≤ 380 N total; tip SF ≥ 2.09 and sliding SF ≥ 1.28 at the full resultant.

Pelvis lift: 110.4–126.1 N at 300 N (≤ 171 N under passive slip), below the 294 N cap (0.5 body weight of 60 kg).

Statics only; outriggers deployed, mu = 0.5, rigid frame.

Source: team statics analysis of design D3. Unbuilt and unvalidated.

A7 · Passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total



Group-tuned thresholds (corner 150, lateral 155, centre 120 N): ≥ 300 N along the fall (305 N minimum over 360 motion directions, nominal pelvis), worst-case horizontal resultant 380 N, below the 420 N static envelope (SF 1.5; SF 1.0 at 640 N). The perpendicular component reaches 178 N (28° off the fall); independent slipping does not preserve force direction. Tip SF ≥ 2.09, sliding SF ≥ 1.28 at the full resultant; the walker slides at 465 N (259 N if mu = 0.3).

Source: team statics analysis of design D3. Unbuilt and unvalidated.

A8 · Arrest coverage and lock-delay sensitivity

Lock delay	Caught trials	D3, passive slip brakes (headline)	D3, fixed 300 N along the fall (old metric)	Original design, fixed 300 N
0.28 s (baseline)	814	805/814 = 98.8% (CI 97.9–99.5)	804/814 = 98.7% (CI 97.7–99.4)	377/814 = 46.3% (CI 42.9–49.8)
0.15 s	844	821/844 = 97.2% (CI 95.9–98.2)	820/844 = 97.1% (CI 95.8–98.1)	170/844 = 20.1% (CI 17.5–23.0)
0.08 s	866	817/866 = 94.3% (CI 92.5–95.7)	807/866 = 93.1% (CI 91.3–94.7)	56/866 = 6.4% (CI 5.0–8.4)

Headline coverage uses the force the passive slip brakes actually produce for each trial's fall direction: the along-fall component in the stopping model, the full resultant in the static-tip, slide and dynamic-tip checks. Arrested = within 0.25 m, no dynamic tip, static tip SF ≥ 1.5, slide ≤ 1 cm. Wilson 95% CI.

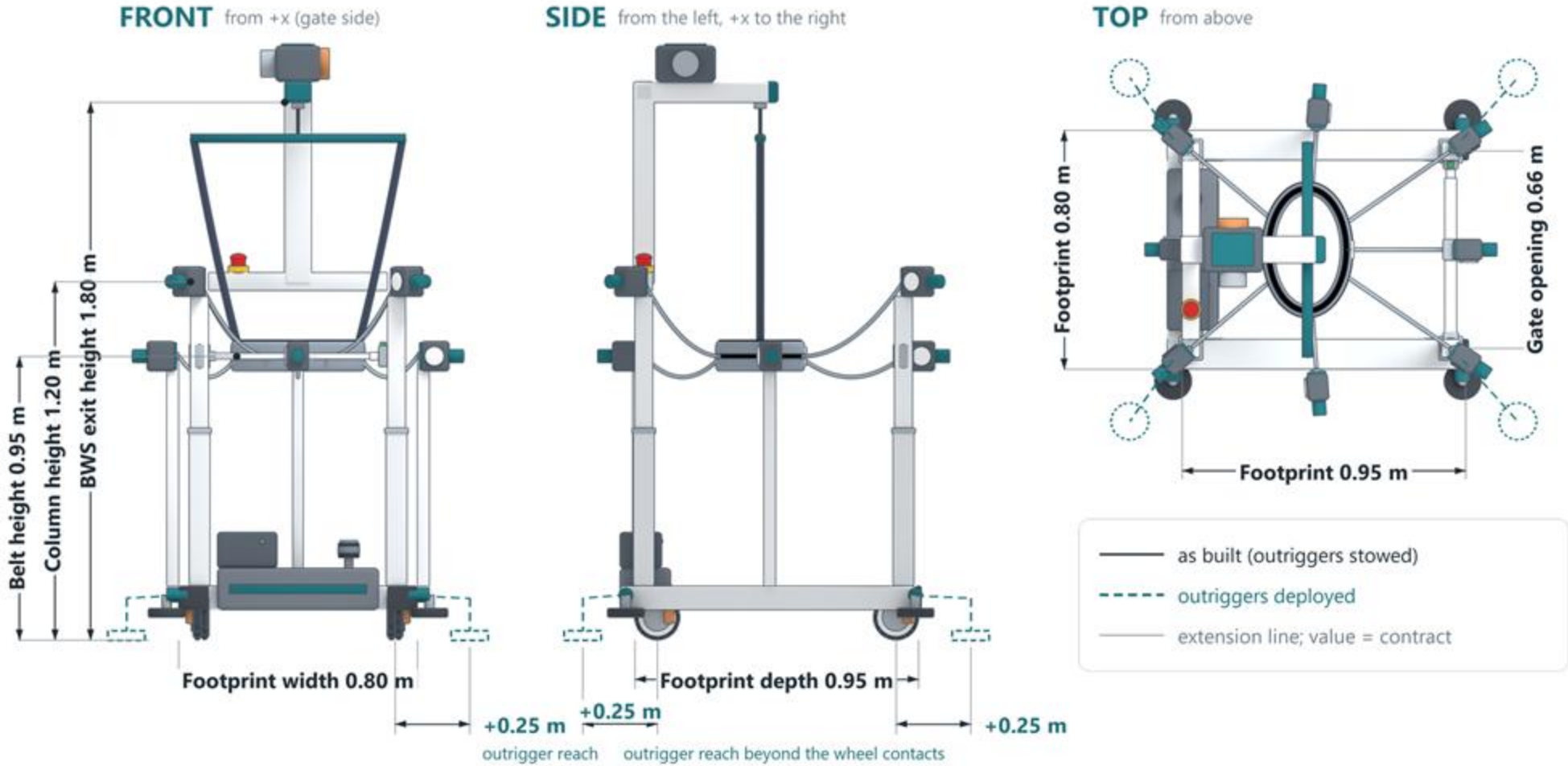
Old metric, lock 0.28 s, fixed along-fall force of 200 / 300 / 400 N: 752 / 804 / 806 of 814. Allowing 491 N is worse: 775/814 = 95.2%, because the static SF-1.5 check fails above 420 N.

Baseline falls 95/2000; the 9 non-arrested caught trials could add to that: 104/2000 at worst. Not a fall-reduction claim.

COM stopped a median 0.104 s after engagement (p95 0.19 s; descriptive, nearest), 814 trials at lock 0.28 s. Slip-brake energy ≤ 22.2 J.

Source: team statics and simulation analysis of design D3. One seeded 2,000-trial run; empirical and conditional on the D3 assumptions.

A9 · Design D3: orthographic views and key dimensions



Belt at nominal position, cables slack; outriggers drawn deployed (dashed). Dimensions come from the shared D3 contract. Concept render: unbuilt and unvalidated.

Render: Blender (design D3 CAD).

A10 · Design D3 specification: geometry, spool, brake and servo

Geometry (from the D3 contract)	Value
Arrest force (horizontal)	passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total (305 N minimum over 360 directions)
Static tip	SF ≥ 1.5 required; envelope 420 N (SF 1.5), 640 N (SF 1.0); SF ≥ 2.09 at the full slip resultant
Walker mass and CG	80 kg incl. ~20 kg low ballast; CG (0.45, 0, 0.4) m
Wheel contacts	x = 0.075 / 0.875, y = ±0.325 m (footprint 0.95 × 0.8 m)
Outrigger feet	+0.25 m all round, deployed on catch; rubber pads, mu ≥ 0.5
Anchors (cable exits)	4 corner at z = 1.20 m; 2 lateral and 2 centre at z = 0.95 m
BWS anchor (cantilever)	(0.42, 0, 1.8) m above the pelvis; hang load 1251 N
Pelvis and belt	pelvis (0.42, 0, 0.95) m; belt half-depth 0.13 m, half-width 0.205 m
Front gate arm	40×20×2 mm 6061-T6 over 0.66 m, latched; cable 150 N, yield SF 4.68; arms only when latched

Spool, brake and servo	Value (basis)
Passive slip thresholds	corner 150 N, lateral 155 N, centre 120 N (spring-applied, power-off brakes)
Slip torque at the spool	3.75 / 3.88 / 3.00 N·m (threshold × spool radius 25 mm)
Rated hold torque	5.63 / 5.82 / 4.50 N·m (1.5 × slip torque)
Worst cable tension, 300 N arrest	199.6 N (controller-set case, worst single cable, 24 directions)
Nudge	150 N resultant, T_max 101.1 N per cable; servo torque 2.53 N·m at the spool; tip SF ≥ 4.36
BWS spool brake	46.91 N·m; hang load 1251 N (1.5 × 85 kg × g, overhead, no horizontal load)
Slip-brake energy	≤ 22.2 J (16.7 J at 0.28 s; arrested trials, three lock delays)
Pelvis lift caps	arrest 294 N (≤ 171 N under passive slip); nudge 88 N (0.5 BW and 0.15 BW of 60 kg)

Not validated: no stress analysis, spool inertia or backlash; mu, cable stiffness and servo data are assumptions. Geometry comes from one contract shared by the CAD and the analysis (generated CAD parts are checked against it to 1 mm).

Source: team statics analysis of design D3 (x forward, y left, z up).

A11 · Closest prior work and how CatchPoint differs

Work	What it does	How CatchPoint differs
RobUST (Omofuma 2023)	Cable pelvis robot; restoring force outside a fixed, subject-specific region	Predictive capture point, step-reach region, escalation to a catch
MRBA / DRBA (NTU)	Pelvic robots; sensor thresholds trigger a brake lock	Preserves the patient's own recovery step; graded force
ExoRecovery (Orhan 2023)	Exoskeleton plans and executes the recovery step	Patient keeps stepping; robot only helps
Monaco APO (2017)	Hip exoskeleton torque once a deviation is detected	Walker interface; recoverable-region logic
mTPAD (Stramel 2023)	Cable pelvic robot on a rollator, about 35 N per cable	Recoverability-based control; conceptual slip-limited catch mode

"To our knowledge": no work found that combines all four ingredients; absence of evidence from a search is not proof.

Sources: references [1]–[6]. Some entries were read from abstracts or summaries only.

A12 · Physics-demo protocol, calibration, and limits

Setting	Falls	Catch-lock	Nudged	Hands-off
No robot	32	–	–	24
B1 θ = 0.08 m	0	46	–	10
B1 θ = 0.11 m	0	41	–	15
B1 θ = 0.14 m	0	37	–	19
B1 θ = 0.18 m	1	31	–	24
CP r0 = 0.10 m	0	24	27	5
CP r0 = 0.14 m (default)	0	26	23	7
CP r0 = 0.20 m	0	33	14	9
CP r0 = 0.28 m	0	33	9	14

- 7 push magnitudes (60–300 N) × 8 gait phases; all pushes toward the weak side; forces mass-scaled to a 75 kg equivalent.
- B1 thresholds: θ=.08/.11/.14/.18 m. CP regions: r0=.10/.14/.20/.28 m. Both use the same delays, 0.28 s lock, cable model, and harness catch.
- PPO training used 9.6M steps of unassisted stochastic pushes; the same 56-grid test reduced B0 falls 32→3, with no held-out evaluation.

Kinematic walker; no cable/spool dynamics or collisions; harness torque is assumed; 'caught' denotes simulated catch-lock engagement.

Source: MuJoCo physics demo, Unitree G1 policy (calibrated XCoM region, residual-based escalation).

A13 · Walking demo (v3): held-out test and no-push walks

Pushed while walking	Alone	Nudged	Caught	Fell	Impulse N·s
Healthy, no robot	46	0	0	2	0
Patient, no robot	31	0	0	17	0
Threshold lock	31	0	12	5	20
Always-on spring	0	47	1	0	620
CatchPoint	8	33	7	0	87

No pushes	Walks with a fall	Locks	Walked (m)
Healthy, no robot	0/20	0	8.7
Patient, no robot	13/20	0	9.0
Threshold lock	13/20	0	9.0
Always-on spring	0/20	15	0.8
CatchPoint	9/20	1	9.4

Held out: settings fixed on even push timings × 100–250 N; tested on 48 pushes at odd timings, incl. 125 and 225 N never used for tuning.

Patient: right-leg torque ×0.6, 40 ms delay, walking at 0.3 m/s. Impulse: mean device impulse per push, 75 kg-equivalent.

0/48 falls is compatible with a true rate up to about 6% (95%).

20 walks × 30 s, stronger patient: torque ×0.5, 60 ms delay, stiff knee, motor noise.

Controllers not re-tuned. Falls are mostly forward trips, which CatchPoint was not tuned for; the spring avoids falls but the patient barely walks.

Source: MuJoCo physics demo v3, Unitree G1 policy; one robot, one push direction, deterministic. Robot-policy demonstration, not patient evidence.

A14 · References and image credits

[1] Mobile Robotic Balance Assistant (MRBA), J. NeuroEng. Rehabil., 2023.

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[6] D. M. Stramel et al., "Overground robotic gait trainer mTPAD improves gait symmetry and weight bearing in stroke survivors," Bioengineering, 2023.

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[11] World Health Organization, "Falls," fact sheet.

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